

# Age/size effects on juvenile green sturgeon, *Acipenser medirostris*, oxygen consumption, growth, and osmoregulation in saline environments

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**Abstract** The green sturgeon, *Acipenser medirostris*, is an anadromous species that migrates from freshwater (FW) to seawater (SW) relatively early in its life history, although the ages and sizes of juveniles at SW entry are not known. Developmental constraints of osmoregulatory organs may either prohibit (i.e., due to salinity tolerance limits) or minimize (i.e., due to substantial osmoregulatory or ionoregulatory energetic costs) SW entry in small fish. Interestingly, larger green sturgeon are often encountered in brackish water (BW) estuaries, perhaps due to an energetic advantage in occupying these near-isosmotic environments. To test hypotheses concerning fish-size effects on the energetic costs of occupying habitats of different salinities, we measured oxygen consumption rates in green sturgeon representing three age groups (100, 170, and 533 days post hatch; dph), which were acclimated for 5 weeks to one of three salinities (FW, <3‰; BW, 10‰; or SW, 33‰). Also, after 7 weeks, final wet masses were compared and blood and muscle tissue samples were taken to assess osmoregulatory abilities. There were no differences in body-mass-adjusted oxygen consumption rates between

any salinities or ages, indicating that the energetic costs were not prohibitively high to occupy any of these salinities. The only mortalities occurred in the 100 dph SW group, where 23% of the fish died, from apparent starvation. Final wet masses were comparable between FW and BW for each age group and with the 533 dph SW group, but were lower in SW groups at 100 and 170 dph. Similarly, osmoregulatory abilities, in terms of plasma osmolality, Na<sup>+</sup>, K<sup>+</sup>, lactate, and protein concentrations, and muscle water content, were comparable in FW and BW groups at all ages, and with the SW group at 533 dph. These results indicated an age/body size effect in hyperosmotic adaptability, and that juvenile green sturgeon may be found in FW or BW at any age, but only have the ability to enter SW by 1.5 years (75 cm, 1.5 kg) of age.

**Keywords** Sturgeon · Salinity · Metabolism · Osmoregulation · Growth

## Introduction

Sturgeons are large, migratory fishes that occur in major river systems of the northern hemisphere (Bemis and Kynard 1997). Many sturgeon species are semi-anadromous, spending a significant portion of their life in brackish water (BW) habitats, and a few are truly anadromous, spending a

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significant portion of their life in full-strength oceanic seawater (SW) (Doroshov 1985). Many sturgeons are of cultural (Holzkamm and Waisberg 2004), ecological and economic importance, although most species are also in decline (Birstein 1993). Sturgeons are also very interesting for their evolutionary relationships, representing the ancestral line of bony fishes (Chondrostei). Despite this, many aspects of their early life history, when fish are known to be the most susceptible to environmental perturbations and predation (Sogard 1997), are not well understood (McCormick et al. 1997). In particular, the timing of juvenile migratory movements and SW entry, which are critical life events, are not well known.

Green sturgeon, *Acipenser medirostris*, are anadromous, culturally important to Native Americans, and have supported commercial and recreational fisheries. There is some evidence populations are in decline, and the Sacramento River population has recently been listed/proposed for listing as threatened.<sup>1</sup> These fish are known to reproduce in just three river systems along the Pacific Coast of North America: the Rogue in Oregon, and the Klamath and Sacramento in northern California (Moyle 2002). Green sturgeon may be the youngest sturgeon species to enter full-strength SW. The eggs and larvae are large in comparison to most other sturgeon species and they grow rapidly (Deng et al. 2002), move into BW habitats (Radtke 1966), presumably entering the ocean within the first three years of life.<sup>2</sup> This life history strategy is likely a reflection of some of their natal watersheds, which have high gradient rivers with small estuaries (Artyukhin and Andronov 1990). Although the Sacramento River system consists of a large river with an expansive delta and estuary, which may be one of the reasons that it appears to be a discrete spawning population (Israel et al. 2004). Unfortunately, the findings of field studies

have been limited due to the rarity of this species and the high gradient or high turbidity river systems they occupy.

Salinity adaptation is a key factor in the life history of juvenile green sturgeon and other anadromous fishes. Salinity poses osmotic challenges to organisms, and the energetic cost due to osmoregulation in different salinities can be significant (Boeuf and Payan 2001). Most recent studies on the aerobic metabolic costs of salinity in teleosts have not found large costs (Boeuf and Payan 2001), except when salinities are changing rapidly (Morgan et al. 1997). Although some aerobic metabolic studies have been conducted on sturgeon juveniles at different salinities (Shelukhin et al. 1990; Secor and Gunderson 1998; McKenzie et al. 1999, 2001a), there have been no studies investigating size or age effects on salinity-associated metabolic costs. Metabolism measurements, as physiological tools, can be used to make predictions about organisms' optimal environmental conditions (Cech 1990). For example, some sturgeon may have survived over the millennia because they are either energetically adapted to BW conditions, or are restricted there at relevant life stages. They may be especially useful for predicting distribution of juvenile green sturgeon, which are rarely collected.

Our objective was to answer the question: at what age and size are juvenile green sturgeon physiologically and energetically capable of inhabiting brackish and SW habitats? To answer this question, three age groups of juvenile green sturgeon were acclimated to three salinities, and measured for aerobic metabolism (oxygen consumption), trends in growth (final wet masses), and iono- and osmoregulatory abilities. The salinities were chosen to mimic green sturgeon environments of fresh water (FW), BW, and SW. In addition, the BW salinity was chosen because it is nearly isosmotic with internal osmolality (Altinok et al. 1998; LeBreton and Beamish 1998). We hypothesized that the cost of osmoregulating in SW decreases as the fish get older and larger. We also hypothesized that there would be an energetic advantage to occupying BW, that would be reflected by lower aerobic metabolic rates as compared to either FW- or SW-acclimated fish. Finally, we decided to base our

<sup>1</sup> Biological Review Team. 2005. Green sturgeon (*Acipenser medirostris*) status review update. NOAA Fisheries, Southwest Fisheries Science Center, Santa Cruz, CA. 31 pp.

<sup>2</sup> Nakamoto, R.J., Kisanuki, T.T., Goldsmith, G.H. 1995. Age and growth of Klamath River green sturgeon (*Acipenser medirostris*). pp 20, U.S. Fish and Wildlife Service, Klamath River Fishery Resource Office, Yreka, CA.

determination of when the fish are capable of existing in SW on whether or not they can survive, grow, and regulate water and ions at statistically indistinguishable levels to fish in FW, or similarly to typical ionic increases (5–20%) found in anadromous salmonids in SW (Holmes and Donaldson 1969).

## Methods

### Salinity acclimation

Adult wild green sturgeon from the Klamath River were spawned and hatched according to the methods described by Van Eenennaam et al. (2001). Three age groups (100, 170, and 533 days post hatch; dph) of first hatchery generation juveniles were acclimated to either FW ( $<3\text{‰}$ ), BW ( $10\text{‰}$ ), or SW ( $33\text{‰}$ ). The 100 dph and 170 dph fish were from the 2003 birth cohort, and the 533 dph fish were from the 2002 birth cohort. These ages were chosen to document changes in fish during the period of SW adaptation. Based on a previous experiment, sturgeon at 100 dph struggle to survive in SW, at 170 dph they can survive for up to 28 days in SW, thus, any changes that may be taking place may be captured by these two age groups (Allen 2005). The 533 dph fish served as a much older group to determine if the fish at 170 dph were fully adapted for SW life. For the 100, 170 and 533 dph groups ( $n = 90, 90$ , and  $60$ , respectively), initial mean total lengths (TL;  $\pm$  standard error; SE) were:  $18.3 \pm 0.2$  cm,  $34.3 \pm 0.3$  cm, and  $75.2 \pm 0.7$  cm, and initial mean wet weights ( $\pm$ SE) were:  $23.7 \pm 0.8$  g,  $145.7 \pm 3.8$  g, and  $1539 \pm 46$  g respectively. All fish were initially held at  $0\text{‰}$  prior to experiments, and gradually acclimated to salinities greater than FW at a rate of  $5\text{‰} 24 \text{ h}^{-1}$ , taking SW fish 7 days to reach  $33\text{‰}$ . Because it is not known the duration of time juvenile green sturgeon require to transition into SW, a time duration was chosen where there was no mortality while fish were experiencing increasing salinities. Fish were also acclimated to  $12\text{--}13^{\circ}\text{C}$  (Table 1), the SW temperature, at  $1^{\circ}\text{C d}^{-1}$  from pre-experimental conditions at  $19^{\circ}\text{C}$ , according to Cocking (1959) and Fry (1971). Once the SW treatment group had

reached its final salinity, all fish were held at their treatment salinities for an additional 44–53 days before completion of the experiment.

The 100 dph fish were divided into groups of 30 per treatment, and held at the Center for Aquatic Biology and Aquaculture on the University of California, Davis campus. During salinity and temperature adjustments, fish treatment groups were held in separate 50-l containers that were aerated and situated in temperature controlled water baths. After the SW group reached  $33\text{‰}$ , all fish were transferred to holding tanks. FW fish were held in an insulated 220-l, rectangular tank, with a chilled, flow-through, non-chlorinated, well-water supply. The BW and SW fish were held in separate re-circulating systems, each with a 220-l insulated, rectangular holding tank, a 10-l solids settling tank, an 80-l sump tank with biofilter media, submersible heaters attached to a YSI 72 Proportional Temperature Controller (Yellow Springs Instrument Co., Yellow Springs, Ohio, USA), stainless steel chilling coils,  $30\text{-}\mu\text{m}$  and  $10\text{-}\mu\text{m}$  filters, and a large trickle column. Temperature was monitored every 15 min by submersible temperature loggers. Salinity (salinity refractometer), dissolved oxygen (YSI 55 meter), pH, and ammonia (salicylate method, colorimetric kit) were also measured daily. An ammonia detoxifier (Amquel, Novalek, Inc., Hayward, CA, USA) was also added periodically at manufacturer's recommended dosages to keep ammonia levels low ( $<20 \mu\text{g NH}_3 \text{ l}^{-1}$ ). Water changes representing at least 15% of each system's volume were made daily using filtered SW (diluted to the proper salinity) from the Bodega Marine Laboratory (BML), a coastal laboratory between the Sacramento and Klamath Rivers. The experiment was conducted using a simulated natural photoperiod, and fish were fed commercial diet (Nelson & Sons, Inc., Murray, UT, USA; Silver Cup Trout Diet) at  $2.4\%$  body mass  $\text{d}^{-1}$  according to an optimal growth curve based on body mass and temperature that was derived for white sturgeon, *A. transmontanus* (S. I. Doroshov unpublished data). Fish were fed during crepuscular and nighttime periods using 12-h belt feeders to maximize food consumption when fish were active and searching for food (Kynard et al. 2005). Wastes and uneaten food were removed by siphon twice daily.

**Table 1** Water quality data

| Age (dph)   | Trt | Salinity (‰) | Temp (°C)   | DO (% sat.) | DO (mg l <sup>-1</sup> ) | pH          | NH <sub>3</sub> (μg l <sup>-1</sup> ) |
|-------------|-----|--------------|-------------|-------------|--------------------------|-------------|---------------------------------------|
| 100         | FW  | 0.1 ± 0.02   | 12.9 ± 0.01 | 97.0 ± 0.3  | 10.10 ± 0.07             | 7.75 ± 0.04 | 0.1 ± 0.01                            |
| 100         | BW  | 10.2 ± 0.10  | 12.9 ± 0.01 | 93.1 ± 0.7  | 9.23 ± 0.06              | 8.25 ± 0.02 | 12.5 ± 2.74                           |
| 100         | SW  | 33.4 ± 0.12  | 12.9 ± 0.01 | 92.8 ± 0.5  | 8.07 ± 0.04              | 7.86 ± 0.03 | 4.0 ± 1.10                            |
| 170 and 533 | FW  | 2.7 ± 0.01   | 12.0 ± 0.01 | 88.1 ± 0.8  | 9.36 ± 0.08              | 7.61 ± 0.01 | 17.3 ± 0.49                           |
| 170 and 533 | BW  | 9.9 ± 0.07   | 11.9 ± 0.01 | 87.6 ± 1.1  | 8.89 ± 0.10              | 7.73 ± 0.01 | 19.4 ± 0.37                           |
| 170 and 533 | SW  | 33.2 ± 0.03  | 12.2 ± 0.01 | 90.4 ± 1.0  | 7.90 ± 0.08              | 8.00 ± 0.01 | 1.9 ± 0.03                            |

Data represent means ± SE. Abbreviations: dph = days post-hatch, Trt = Treatment, DO = dissolved oxygen, sat. = saturation

The 170 and 533 dph fish were divided into groups of 30 and 20 fish respectively, and held at the BML. These fish were brought from 0‰ CABA holding tanks, and transferred to <3‰ BML holding tanks at the onset of the experiment, where salinity and temperature were adjusted as previously described. Both age groups were held together in the same tanks in each of the three salinities in 2-m diameter, 2000-l circular fiberglass tanks. The FW tank received flow-through chilled well-water, the SW tank received flow-through ambient SW, and the BW tank received a mixture of the two. Temperature was monitored every 15 min by submersible data loggers. Salinity, dissolved oxygen (both measured using a YSI 85), pH, and ammonia were also measured daily (Table 1). All tanks were on a natural photoperiod, and fish were fed commercial diet (Nelson & Sons, Inc., Murray, UT, USA; Silver Cup Trout Diet) at 1.54% body mass d<sup>-1</sup> for 170 dph fish and 0.59% body weight d<sup>-1</sup> for 533 dph fish according to the white sturgeon optimal growth curve (S. I. Doroshov unpublished data).

The difference in experimental salinities for the FW treatments (0‰ for 100 dph, and <3‰ for 170 dph and 533 dph) was not considered to be a factor. These salinities were both hypo-osmotic to the fish, and at the completion of the experiment there was no difference in plasma osmolality, Na<sup>+</sup>, or Cl<sup>-</sup>, although there was a slightly lower plasma K<sup>+</sup> concentration (2.1 ± 0.1 mEq l<sup>-1</sup> vs. 3.6 ± 0.2 mEq l<sup>-1</sup> and 3.2 ± 0.1 mEq l<sup>-1</sup>) in 0‰ (based on analysis of variance; ANOVA).

For both locations, during daily system maintenance, fish health was monitored. External assessments of good fish health were based on

observations of food consumption, slow to moderate gill ventilation movements, demersal swimming behaviors, and no exterior physical signs of stress. Unhealthy fish were determined to be those that were not consuming food, exhibited rapid or very slow gill ventilation movements over a prolonged period of time, had abnormal swimming behaviors such as swimming at the surface or continuous swimming, appeared to be dehydrated or emaciated, and sometimes exhibited a reddening of the vasculature and tissues around the rostrum and fins.

#### Aerobic metabolism

Resting aerobic metabolism was determined by oxygen consumption using flow-through, clear, acrylic respirometers. Cylindrical, 4-l respirometers were used for the 100 dph group, and triangular cross-section 6.5-l or 26-l respirometers were used for the 170 dph and 533 dph groups respectively. For the 100-dph group, six respirometers were run at one time, with one blank respirometer per run to test for microbial respiration rates (Cech 1990). For the 170 and 533 dph groups, three respirometers of each type were run at one time. To account for the maximum amount of microbial respiration at each salinity, the same respirometers were used as blanks after fish were removed on one of the final experimental days. Small corrections were made in the 100 dph group data to account for microbial respiration. These respirometers were also soaked in dilute bleach solution and rinsed thoroughly between measurement periods (Cech 1990) whenever any microbial respiration was detected. No corrections were needed for the 170-dph and 533-dph groups. All respirometers were held at constant

temperatures by surrounding water baths, and were visually isolated from one another by either barriers or separate water bath tanks. Oxygen consumption was measured at 35–41 days after reaching the treatment salinities. Post-absorptive (fasted 72 h) fish were placed into respirometers by 1930 h, and the tanks were covered with shade cloth for overnight acclimation. Water flow rates through the respirometers were reduced to experimental flow rates after 60 min for the 100 dph fish, and at 0600 the next morning for 170 dph and 533 dph fish. Water samples collected for oxygen partial pressure ( $pO_2$ ) measurements were always given sufficient lag time (Steffensen 1989), after any flow rate adjustments. Water samples were collected using syringes and Tygon microbore plastic tubing (wall thickness: 0.76 mm, Norton Performance Plastics Corp., Akron, OH, USA) from aerated head tanks and outflowing respirometer tubes to measure inflowing and outflowing  $pO_2$ s, respectively. The  $pO_2$ s were determined using a polarographic oxygen electrode (Model E101, Analytical Sensors, Inc., Sugarland, TX, USA) fitted into a water-thermostatted cuvette that was connected to a Radiometer PHM71 Mk2 Acid-Base Analyzer (Radiometer America, Cleveland, OH, USA). Oxygen content was determined using a nomogram (Green and Carritt 1967). Tail beat frequency (TBF) and ventilation frequency (VF) were visually estimated from fish prior to sampling, and recorded from each fish over a 30-s period immediately after water sample collection. Fish were determined to be in a resting state if TBF was  $\leq 15$  beats  $\text{min}^{-1}$ , although fish were also included if TBF was higher but either of very low amplitude or if VF was low ( $\leq 70$  ventilations  $\text{min}^{-1}$ ), based on measurements and observations taken during pilot studies. Some of the younger fish tended to have frequent, small body movements, an observation that has also been made in juvenile white sturgeon (Crocker and Cech 1997). Fish with  $< 2$  resting measurements of oxygen consumption were excluded from analyses. For the 100 dph group, measurements for 25–30 fish per salinity were conducted, and a total of 20 fish were excluded. For the 170 dph and 533 dph group, 12 fish were measured for each salinity and age combination, and a total of seven fish were

excluded. Body-mass-adjusted oxygen consumption measurements ( $MO_2$  in  $\text{mg O}_2 \text{ h}^{-1} \text{ kg}^{-1}$ ) were determined from the following equation:

$$MO_2 = (O_{2\text{in}} - O_{2\text{out}}) \cdot (V_w \cdot 60) \cdot M_b^{-1} \quad (1)$$

where  $O_{2\text{in}}$  = inflowing water oxygen content ( $\text{mg l}^{-1}$ ),  $O_{2\text{out}}$  = outflowing water oxygen content ( $\text{mg l}^{-1}$ ),  $V_w$  = water flow rate ( $\text{l min}^{-1}$ ), and  $M_b$  = body mass (kg). The 1.0 body mass scaling coefficient was chosen because it closely approximated the slope derived from the  $\log_{10} MO_2$  ( $\text{mg O}_2 \text{ h}^{-1}$ ) versus  $\log_{10}$  body mass (kg) relationship for green sturgeon in this experiment for all salinities and ages ( $y = 0.98x - 1.79$ ,  $R^2 = 0.94$ ) and those derived by Mayfield and Cech (2004).

### Survival and growth

As a part of the experiment, sturgeon wet masses and mortalities were recorded. Mortalities were removed promptly from tanks, and were not used for any blood or tissue analyses, but were included in counts of survival at the termination of the experiment. Growth analyses were an evaluation of trends seen in different age and salinity groups, because sufficient replication of tanks was not possible for formal statistical analyses. For analyses of growth trends, all fish were fasted for 24 h ( $19^\circ\text{C}$ ) at the start of the experiment, and for 72 h ( $12\text{--}13^\circ\text{C}$ ) at the termination of the growth phase of the experiment (42 d in SW, experiment day 48). After fasting, fish were temporarily anesthetized in 100  $\text{mg l}^{-1}$  tricaine methanesulfonate (MS-222) that was dissolved in BW (10‰) at the initiation of the experiment, or at their treatment salinities at the end of the experiment, and weighed and measured for TL. At the start of the experiment, fish were then randomly divided into treatment groups, and at the end of the growth study, normal ration levels were resumed until the completion of the experiment. Specific growth rates were calculated for tank means according to Ricker (1979) and Wurtsbaugh and Cech (1983).

### Blood and muscle measurements

At the completion of the experiment (44–53 d in SW), all fish were overdosed in 500  $\text{mg l}^{-1}$  MS-

222 at their treatment salinity, and then quickly (<5 min) weighed, measured, and sampled for blood and muscle tissue. Blood samples were collected from the 100 dph group by either cardiac puncture using heparinized, 1-ml syringes with a 23 Ga hypodermic needles, or by severing the caudal fin and collecting blood from the caudal vein using heparinized microhematocrit capillary tubes in the smallest fish (<10 g). In the 170 dph and 533 dph groups, cardiac puncture was used to collect blood using 20 Ga or 18 Ga needles, respectively. After collection, blood samples were gently agitated by hand, and a hematocrit sample was collected in a heparinized microhematocrit capillary tube. Microhematocrit tubes were then centrifuged at  $12,600 \times g$  for 3 min, and blood samples were centrifuged at  $1500 \times g$  for 5 min. Plasma was then collected in 0.5–4.0 ml cryovials, snap frozen in liquid nitrogen, and stored at  $-80^{\circ}\text{C}$  until analysis (<2 mo). Gill, kidney, pyloric caeca and intestinal tissue were removed for  $\text{Na}^+$ ,  $\text{K}^+$ -ATPase activity analysis, and are the subject of another paper (Allen 2005).

Plasma ( $n = 19\text{--}30$ ) was measured for osmolality,  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{K}^+$ , glucose, lactate, and protein concentrations. Plasma osmolality was measured using a Vapro 5520 Vapor Pressure Osmometer (Wescor, Inc., Logan, UT, USA). Plasma sodium ( $\text{Na}^+$ ) and potassium ( $\text{K}^+$ ) were measured using an IL 343 Digital Flame Photometer (Instrument Laboratory, Inc., Lexington, MA, USA). Plasma chloride ( $\text{Cl}^-$ ) was measured on a Radiometer CMT10 Chloride Titrator (Radiometer America, Cleveland, OH, USA). Plasma glucose and lactate were measured on a YSI 2700 Biochemistry Analyzer. Plasma proteins were measured according to “Doumas et al. (1981)” using Sigma Diagnostics Total Protein Reagent 541-2 (Sigma Diagnostics, Inc., St. Louis, MO, USA), in an Aquamate Spectrophotometer (Thermo Spectronic, Rochester, NY, USA). All plasma tests were run in triplicate whenever possible, depending on plasma volume, except for plasma protein analyses, which were run in duplicate.

A 1–3 g sample (depending on fish size) of epaxial musculature approximately 3 scutes posterior to the head, was removed from each fish for muscle water content analysis. Muscle samples

were placed on ice and then stored at  $-20^{\circ}\text{C}$  for <7 d prior to analysis. Samples were initially weighed and then dried at  $60^{\circ}\text{C}$  on individual aluminum foil squares until no further weight loss occurred (14 d). Muscle water content was determined according to the equation:

$$\text{Muscle water \%} = ((W_m - D_m) \cdot W_m^{-1}) \cdot 100 \quad (2)$$

where  $W_m$  = wet mass (g) and  $D_m$  = dry mass (g).

### Statistical analyses

Two-way ANOVA's (individual fish as replications) were used to compare factors of initial age and salinity for mass-adjusted aerobic metabolic rate; plasma osmolality,  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{K}^+$ , lactate, and protein; and hematocrit and muscle water content (SAS Institute, NC, USA; JMPIN 4.0.4 academic version 2001). If the interaction between age and salinity was found to be significant, individual treatment groups were compared. Tukey's honestly significant difference multiple comparison tests of means were used only if the overall ANOVA was found to be significant. Data were  $\log_{10}$  or square root transformed to meet ANOVA assumptions when necessary. Plasma glucose data were analyzed by Kruskal–Wallis non-parametric tests, and a Bonferroni correlate (Neter et al. 1996) was used to test for differences between treatment groups within each age. For growth comparisons, mean initial and final masses of salinity treatment within each age group (each mean from within one tank) were compared using student's  $t$ -tests. Specific growth rates were calculated for tank means only if there was a significant difference between mean initial and final mass. Growth analyses do not represent formal statistical tests, as tank replication ( $n = 1$ ) was not sufficient, but was conducted to show growth trends. Student's  $t$ -tests were used to compare plasma osmolality,  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{K}^+$ , glucose, lactate, and protein; and muscle water content between SW 100 dph fish that appeared healthy or unhealthy at the end of the experiment. A Pearson product–moment correlation coefficient was used to test the correlation between osmolality and wet mass in the 100 dph SW group. Linear regressions were used to describe the relationships between

the log of aerobic metabolic rate and the log of body mass, aerobic metabolic rate and TL, and wet mass and TL. In all cases, differences were considered significant at  $P < 0.05$ . With the exception of the regression equations, all statistical values are reported as means  $\pm$  SE.

## Results

### Aerobic metabolism

There were no  $\text{MO}_2$  differences among the different age or salinity groups of juvenile green sturgeon (Table 2), after  $\text{MO}_2$  was corrected for body mass ( $M_b$ ). Sturgeon did consume more oxygen with increasing body mass according to the equation:  $\text{MO}_2 = 61.01 \cdot M_b^{0.98 \pm 0.005}$  ( $R^2 = 0.94$ ) with  $\text{MO}_2$  in  $\text{mg O}_2 \cdot \text{hr}^{-1}$  and body mass in kg, where the slope is presented with a 95% confidence interval. Similarly, sturgeon consumed more oxygen with increasing TL according to the equation:  $\text{MO}_2 = 0.00032 \cdot \text{TL}^{2.91 \pm 0.018}$  ( $R^2 = 0.92$ ) with TL in cm. Fish mass (g) increased with TL (cm) according to the equation:  $\text{Mass} = 0.00420 \cdot \text{TL}^{2.961 \pm 0.011}$  ( $R^2 = 0.97$ ).

### Survival and growth

In general, the youngest green sturgeon adapted the least well to SW. In the 100 dph SW group, 7 out of 30 fish died (23% mortality rate; Table 2). The only other, incidental, mortality occurred in the 533 dph SW group, where one fish jumped out of its tank. Regarding growth trends, the 100 dph and 170 dph FW and BW groups had greater final tank masses than the 100 dph and 170 dph SW groups, respectively (Table 3). At both of these ages, FW and BW groups showed increases in mass whereas the SW groups did not. In contrast, the 533 dph FW and BW groups did not show any growth trend differences as compared to the SW group.

### Osmoregulation and ionic regulation

Plasma osmolality, a measure of overall osmoregulatory ability, was highest in SW at 100 dph and decreased with increasing age and size

(indicating better regulation), with no significant differences among SW, FW and BW treatments detected in the 533 dph age group (Fig. 1A). The FW and BW groups were not different within the same age, and only the 100 dph BW group was elevated over 170 dph FW and BW groups.

In general, the regulation of plasma electrolytes followed age and salinity relationship patterns similar to that of plasma osmolality. Plasma  $\text{Na}^+$  was highest in the SW 100 dph groups and decreased with age, with no significant differences among the SW, FW or BW groups at 533 dph (Fig. 1B). None of the FW or BW treatments were different from each other in all age groups. Plasma  $\text{Cl}^-$  was also highest in the SW 100 dph fish and decreased with age (Fig. 1C). However, the SW treatments had significantly higher plasma  $\text{Cl}^-$  than FW and BW groups at each age, including 533 dph. Plasma  $\text{Cl}^-$  in FW and BW treatments was not different, with the exception of the BW 100 dph group which was elevated over all FW groups. Plasma  $\text{K}^+$  showed a similar pattern across treatments and ages, with high concentrations in the SW treatments (Fig. 1D).

Plasma glucose concentrations were lower in SW treatments for both the 100 dph and 170 dph groups, but were not different from FW or BW groups at 533 dph (Table 2). Similarly, plasma lactate decreased in the SW treatments in the 100 dph and 170 dph age groups compared to FW and BW treatments, although the SW treatment in the 533 dph group was not different from FW and BW treatments at that age (Table 2). The FW and BW treatments were comparable within and among ages, except FW 533 dph, which was elevated compared to younger groups. Plasma protein concentrations showed similar trends to those of plasma lactate (Table 2).

Hematocrit showed no relationships with age and salinity except that the SW (100 dph and 533 dph) groups were both significantly lower than the FW 533 dph group (Table 2).

The changes in muscle water content reflected the sturgeon plasma  $\text{Na}^+$  and  $\text{Cl}^-$  concentrations in the different salinities. At 100 dph and 170 dph, the SW treatment's muscle water content was significantly higher than those in either the FW or BW treatments, with the FW 100 dph group significantly lower than any other group. In

**Table 2** Juvenile green sturgeon aerobic metabolic rate ( $\text{MO}_2$ ), survival, and blood constituents with age (days post hatch) and salinity (‰)

|                                                                   | 100 dph                        |                                |                             | 170 dph                      |                               |                              | 533 dph                       |                                 |                                |
|-------------------------------------------------------------------|--------------------------------|--------------------------------|-----------------------------|------------------------------|-------------------------------|------------------------------|-------------------------------|---------------------------------|--------------------------------|
|                                                                   | 0‰                             | 10‰                            | 33‰                         | 0‰                           | 10‰                           | 33‰                          | 0‰                            | 10‰                             | 33‰                            |
| $\text{MO}_2$ ( $\text{mg O}_2 \text{ hr}^{-1} \text{ kg}^{-1}$ ) | $72.90 \pm 7.33$<br>(20)       | $76.06 \pm 7.63$<br>(18)       | $68.15 \pm 5.29$<br>(21)    | $63.36 \pm 6.02$<br>(10)     | $61.78 \pm 4.65$<br>(10)      | $63.77 \pm 4.51$<br>(11)     | $69.36 \pm 6.98$<br>(12)      | $62.40 \pm 7.41$<br>(12)        | $71.05 \pm 10.99$<br>(11)      |
| % Survival                                                        | 100                            | 100                            | 77                          | 100                          | 100                           | 100                          | 100                           | 100                             | 95 <sup>a</sup>                |
| Plasma glucose ( $\text{g l}^{-1}$ )                              | $0.807^b \pm 0.056$<br>(30)    | $0.615^{ab} \pm 0.053$<br>(26) | $0.672^a \pm 0.216$<br>(22) | $1.222^b \pm 0.126$<br>(29)  | $0.904^b \pm 0.104$<br>(30)   | $0.630^a \pm 0.086$<br>(30)  | $1.353 \pm 0.188$<br>(19)     | $1.248 \pm 0.096$<br>(19)       | $1.206 \pm 0.136$<br>(19)      |
| Plasma lactate ( $\text{g l}^{-1}$ )                              | $0.102^{bc} \pm 0.007$<br>(30) | $0.080^b \pm 0.008$<br>(26)    | $0.036^a \pm 0.007$<br>(22) | $0.088^b \pm 0.006$<br>(29)  | $0.077^b \pm 0.007$<br>(30)   | $0.036^a \pm 0.005$<br>(30)  | $0.107^d \pm 0.007$<br>(19)   | $0.136^{bcd} \pm 0.007$<br>(19) | $0.131^{cd} \pm 0.006$<br>(19) |
| Plasma protein ( $\text{mg dl}^{-1}$ )                            | $2.72^d \pm 0.15$<br>(30)      | $2.52^{cd} \pm 0.25$<br>(29)   | $1.12^a \pm 0.24$<br>(22)   | $2.50^{cd} \pm 0.21$<br>(29) | $1.93^{abc} \pm 0.13$<br>(30) | $1.47^{ab} \pm 0.24$<br>(30) | $2.33^{bcd} \pm 0.10$<br>(20) | $2.12^{bcd} \pm 0.10$<br>(19)   | $2.89^d \pm 0.16$<br>(19)      |
| Hct %                                                             | $22.4^{ab} \pm 0.6$<br>(30)    | $22.0^{ab} \pm 0.6$<br>(29)    | $19.7^b \pm 0.8$<br>(23)    | $20.4^{ab} \pm 0.5$<br>(30)  | $20.9^{ab} \pm 0.5$<br>(30)   | $21.2^{ab} \pm 0.6$<br>(30)  | $23.0^a \pm 0.6$<br>(20)      | $22.7^{ab} \pm 0.8$<br>(19)     | $19.4^b \pm 0.9$<br>(18)       |

Data are means ( $\pm$ SE), except for survival data, with number of individuals measured in parentheses. For  $\text{MO}_2$ , there were no significant differences between ages or salinities, for other blood constituents different letters indicate significant differences (ANOVA); plasma glucose analyzed by Kruskal–Wallis within each age;  $P < 0.05$

<sup>a</sup>The only mortality occurred from one fish jumping out of the tank

**Table 3** Juvenile green sturgeon initial and final masses (mean  $\pm$  SE), and specific growth rates (% body mass (g d<sup>-1</sup>))

| Age (dph) | Trt (‰) | Mean initial mass (g) $\pm$ SE | Mean final mass (g) $\pm$ SE | SGR  |
|-----------|---------|--------------------------------|------------------------------|------|
| 100       | 0       | 23.70 $\pm$ 1.33               | 42.31 $\pm$ 3.21             | 1.21 |
| 100       | 10      | 23.72 $\pm$ 1.37               | 38.72 $\pm$ 4.36             | 1.02 |
| 100       | 33      | 23.59 $\pm$ 1.36               | 24.80 $\pm$ 3.37             | NA   |
| 170       | 2.7     | 139 $\pm$ 6                    | 185 $\pm$ 11                 | 0.62 |
| 170       | 10      | 160 $\pm$ 6                    | 208 $\pm$ 11                 | 0.56 |
| 170       | 33      | 138 $\pm$ 7                    | 130 $\pm$ 9                  | NA   |
| 533       | 2.7     | 1501 $\pm$ 77                  | 1667 $\pm$ 76                | NA   |
| 533       | 10      | 1528 $\pm$ 57                  | 1628 $\pm$ 57                | NA   |
| 533       | 33      | 1587 $\pm$ 102                 | 1632 $\pm$ 97                | NA   |

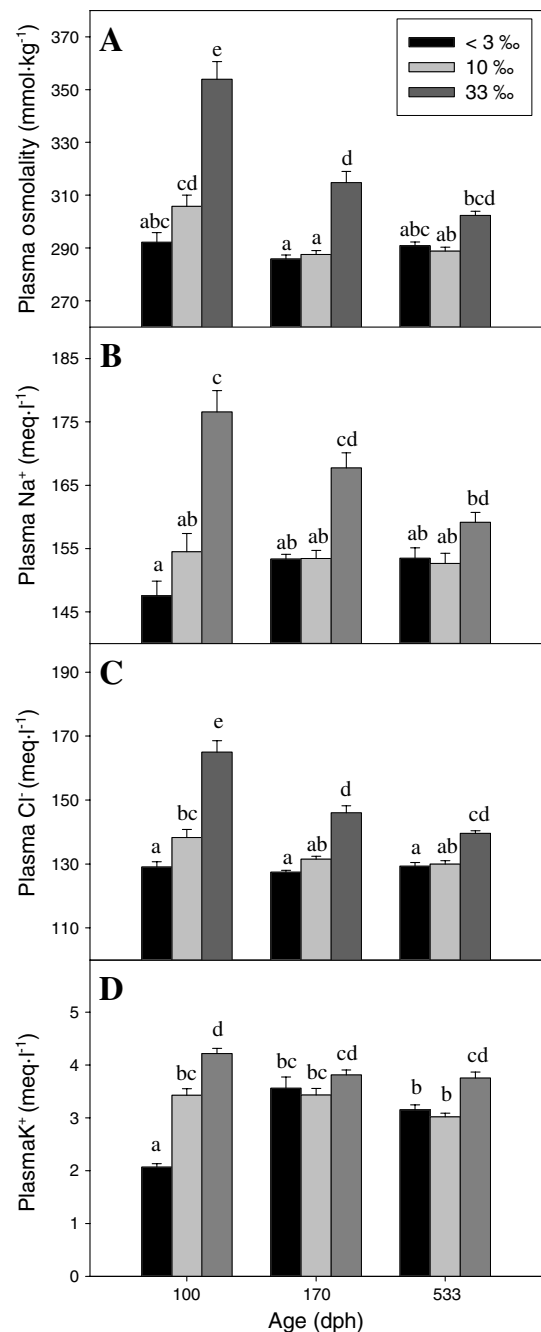
Specific growth rates included only if there was a significant difference between mean initial and final wet mass (*t*-test,  $P < 0.05$ ). In 100 dph and 170 dph fish, there was a trend for greater SGRs in FW and BW than in SW

the 533 dph group, there was no difference between salinity treatments (Fig. 2).

Some fish of the 100 dph group in the SW treatment (6 of 30, 20%) appeared to be in good health based on observations of food consumption and activity levels, compared with others. These fish also had a lower mean plasma osmolality, Na<sup>+</sup>, and muscle water content, and higher mean plasma glucose, lactate, and protein concentrations although they did not differ in plasma Cl<sup>-</sup> and K<sup>+</sup>, and hematocrit. These results indicate a physiologically heterogeneous population, with some fish able to adapt to SW earlier than others at the same age. There was no significant correlation between individual osmolality and final mass for the SW 100 dph group, indicating that age-associated development rather than body size may be a better predictor of successful osmoregulation in SW in juvenile green sturgeon.

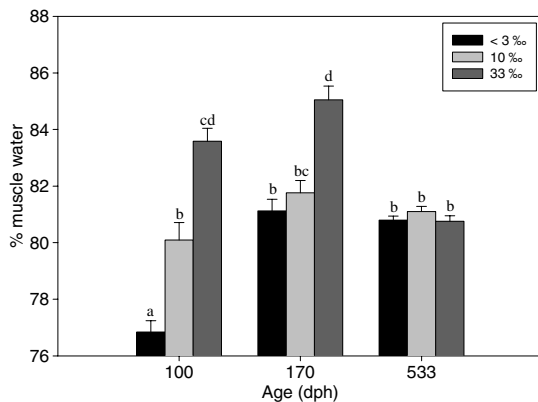
## Discussion

Among sturgeons, green sturgeon appear to be able to successfully enter and osmoregulate in SW at one of the youngest ages. This is evident when taken in context of previous studies carried out on hypo- and hyperosmotic regulation in sturgeons (Table 4). The aerobic metabolic costs associated with these salinity transitions did not change regardless of salinity, indicating that they do not



**Fig. 1** Mean (+SE) juvenile green sturgeon (three different ages, in days post hatch) plasma osmolality (A), Na<sup>+</sup> (B), Cl<sup>-</sup> (C), and K<sup>+</sup> (D), after acclimation to three different salinities (FW, < 3‰; brackish water, 10‰; and SW, 33‰). Different letters indicate significant differences (ANOVA,  $P < 0.05$ ;  $n = 19-30$ )

inhibit juvenile green sturgeon from entering SW. However, iono- and osmoregulation improved with age/size, and juvenile green sturgeon



**Fig. 2** Mean (+SE) juvenile green sturgeon (three different ages, in days post hatch) percent muscle water, after acclimation to three different salinities (fresh water, <3‰; BW, 10‰; and SW, 33‰). Different letters indicate significant differences (ANOVA,  $P < 0.05$ ;  $n = 19$ –30)

appeared most capable of successful osmoregulation at the oldest age tested, 533 dph.

#### Aerobic metabolism

Aerobic metabolic costs were similar in juvenile green sturgeon acclimated to either hyposmotic, isosmotic or hyperosmotic salinities. These results indicate that osmoregulatory costs are not a significant portion of the total energy budget in juvenile green sturgeon. In contrast, studies on tilapia, *Oreochromis mossambicus* (Iwama et al. 1997), and milkfish, *Chanos chanos* (Swanson 1998), have found that oxygen consumption rates can vary even after a long salinity acclimation period. However, similar to our experiments, Morgan and Iwama (1998) found that oxygen consumption did not vary in juvenile coho salmon, *Oncorhynchus kisutch*, acclimated to 0‰, 10‰, and 28‰ despite differences in gill  $\text{Na}^+$ ,  $\text{K}^+$ -ATPase activities. They concluded that either energetic costs due to gill  $\text{Na}^+/\text{K}^+$ -pumps may be small, or that oxygen consumption may not be a good measure of osmoregulatory costs. Recent studies based on  $\text{Na}^+$  transport have determined that costs of osmoregulation may be as low as 4% (Morgan and Iwama 1999) to 10% of the total energy budget (reviewed by Boeuf and Payan 2001). Previous studies on other juvenile sturgeons have revealed similar results to our

findings. Juvenile Adriatic sturgeon, *A. naccarii*, acclimated to 0‰, 11‰, and 23‰ exhibited similar resting metabolic rates (McKenzie et al. 1999), as did juvenile Russian sturgeon, *A. guldenshtadi*, held at 12°C and acclimated to salinities from 0‰ to 12‰ (Shelukhin et al. 1990). However, Jarvis and Ballantyne (2003) suggested that juvenile shortnose sturgeon, *A. brevirostrum*, may have a high cost of osmoregulation in hyperosmotic salinities based on their depressed growth rates (oxygen consumption was not measured).

Rapid salinity changes may result in substantially higher metabolic costs, as has been shown in teleosts (Morgan et al. 1997; Morgan and Iwama 1999). Two studies have found that juvenile sturgeon are able survive in higher salinities if they are acclimated to a moderately hyperosmotic salinity for at least one week (McEnroe and Cech 1985, Altinok et al. 1998). Field data on anadromous juvenile European sturgeons, *A. sturio*, have found that they spend a significant portion of time in estuarine areas prior to moving into more hyperosmotic waters (Rochard et al. 2001). This may indicate that there is an energetic benefit to occupying these nearly isosmotic salinities, however, it may also relate to food abundance and feeding behavior of young sturgeon. Our results did not support this, as there was no clear energetic benefit at any of the salinities tested. The resting aerobic metabolism rates in our study are somewhat lower than those found previously for similar size green sturgeon at 11°C, possibly due to a shorter acclimation time at 11°C in the previous study (Mayfield and Cech 2004). They are also lower than those found in juvenile Gulf sturgeon, *A. oxyrinchus desotoi*, at 23.5°C (Altinok and Grizzle 2003), and juvenile Adriatic sturgeon at 23°C (McKenzie et al. 1999) probably due to temperature, activity, or species-related differences. Our results for green sturgeon are also lower than the oxygen consumption rates for smaller Russian sturgeon at 0–12‰ and 12°C (Shelukhin et al. 1990), and slightly higher than Siberian sturgeon, *A. baerii*, at 15°C (Nonnotte et al. 1993). Altinok and Grizzle (2003) found that juvenile Gulf sturgeon have higher oxygen consumption rates than other teleosts at the same salinities. Although this suggests that osmoregulation may be a higher proportional cost of the

**Table 4** Plasma osmolality and ion concentrations in sturgeons subjected to hypo- and hyperosmotic salinities

| Type             | Species                      | N/rtt            | Life stage | Size (kg) | Salinity (‰) |       | Duration (d) |       | Osmolality (mosm/kg) |              | Na <sup>+</sup> (meq/l) |              | K <sup>+</sup> (meq/l) |             | Cl <sup>-</sup> (meq/l) |              | Paper                          |
|------------------|------------------------------|------------------|------------|-----------|--------------|-------|--------------|-------|----------------------|--------------|-------------------------|--------------|------------------------|-------------|-------------------------|--------------|--------------------------------|
|                  |                              |                  |            |           | Hypo         | Hyper | Hypo         | Hyper | Hypo                 | Hyper        | Hypo                    | Hyper        | Hypo                   | Hyper       | Hypo                    | Hyper        |                                |
| Truly Anadromous | <i>Acipenser medirostris</i> | 22–29 J; 0.3 yr  | 0.02       |           | 0            | 33    | LT           | 44–52 | 292.2 ± 3.6          | 353.9 ± 6.7  | 147.5 ± 2.3             | 176.6 ± 3.4  | 2.1 ± 0.1              | 4.2 ± 0.1   | 129.1 ± 1.6             | 165.0 ± 3.6  | This paper                     |
|                  | <i>A. medirostris</i>        | 29–30 J; 0.5 yr  | 0.15       |           | 2.7          | 33    | 58           | 47–53 | 285.9 ± 1.4          | 314.8 ± 4.2  | 153.3 ± 0.7             | 167.7 ± 2.4  | 3.6 ± 0.2              | 3.8 ± 0.1   | 127.4 ± 0.6             | 146.1 ± 2.2  | This paper                     |
|                  | <i>A. medirostris</i>        | 19–20 J; 1.5 yr  | 1.54       |           | 2.7          | 33    | 58           | 47–53 | 290.8 ± 1.4          | 302.3 ± 1.5  | 153.5 ± 1.6             | 159.1 ± 1.6  | 3.2 ± 0.1              | 3.8 ± 0.1   | 129.3 ± 1.1             | 139.6 ± 0.9  | This paper                     |
|                  | <i>A. medirostris</i>        | 3 SA-A           | 6.0–9.6    |           | FW           | SW    | 3            | 3     |                      |              | 116.0 ± 1.5             | 145.7 ± 9.7  | 1.7 ± 0.1              | 2.1 ± 0.2   | 104.0 ± 2.0             | 151.3 ± 9.1  | Potts and Rudy (1972)          |
|                  | <i>A. oxyrinchus</i>         | 3 J              | 2          |           | FW           | 30    | 25           | 25    | 235.1 ± 2.9          | 300.0 ± 3.9  | 127.0 ± 4.3             | 146.8 ± 3.0  | 3.22 ± 0.23            | 3.10 ± 1.16 |                         |              | Krayushkina (1998)             |
| Semi-anadromous  | <i>A. oxyrinchus</i>         | 20–23 J          | Unk        |           | FW           | BW    | Unk          | Unk   |                      |              | 150.6 ± 2.0             | 164.9 ± 1.2  | 2.68 ± 0.15            | 2.84 ± 0.12 | 113.3 ± 1.7             | 132.9 ± 1.1  | Magnin (1962)                  |
|                  | <i>A. o. desotoi</i>         | 6–8 J; 1.1 yr    | 0.11–0.7   |           | 0            | 35    | LT           | 4     | 260.7 ± 1.6          | 294.2 ± 2.9  | 135.7 ± 1.2             | 152.0 ± 1.7  | 2.0 ± 0.1              | 3.1 ± 0.1   | 106.9 ± 1.9             | 149.2 ± 1.4  | Altinok et al. (1998)          |
|                  | <i>A. sturio</i>             | 14–39 J-A        | Unk        |           | FW           | BW    | Unk          | Unk   |                      |              | 155.8 ± 11.1            | 163.6 ± 1.5  | 4.3 ± 0.02             | 4.65 ± 0.01 | 119.7 ± 13.2            | 126.4 ± 1.6  | Magnin (1962)                  |
|                  | <i>A. transmontanus</i>      | 10 J             | 0.02–0.06  |           | 0            | 25    | LT           | 0.4   | 245 ± 1              | 410 ± 19     |                         |              |                        |             |                         |              | McEnroe and Cech (1985)        |
|                  | <i>A. transmontanus</i>      | 4–11 J; 1.5–2 yr | ~1         |           | 0            | ~33   | LT           | >60   | 268.3 ± 8.3          | 316.0 ± 11.8 | 140.3 ± 6.9             | 163.2 ± 10.1 | 3.53 ± 0.26            | 3.45 ± 0.46 | 116.6 ± 4.8             | 145.8 ± 8.8  | McEnroe and Cech (1987)        |
| FW               | <i>A. transmontanus</i>      | 7 SA-A           | 2.5–15     |           | 0            | 24    | >24          | >24   | 236.1 ± 7.1          | 248.8 ± 13.5 | 131.6 ± 4.4             | 125.0 ± 8.0  | 2.5 ± 0.7              | 2.6 ± 0.8   | 110.6 ± 3.6             | 122.0 ± 3.0  | McEnroe and Cech (1985)        |
|                  | <i>A. transmontanus</i>      | 4–7 SA-A         | 2.5–15     |           | 0            | 35    | >24          | 3     | 236.1 ± 7.1          | 385.0 ± 18.9 | 131.6 ± 4.4             | 200.8 ± 12.6 | 2.5 ± 0.7              | 3.9 ± 0.5   | 110.6 ± 3.6             | 169.3 ± 11.6 | McEnroe and Cech (1985)        |
|                  | <i>A. transmontanus</i>      | 4–5 SA-A         | Unk        |           | FW           | ~33   | >7           | 21–36 | 243–263              | 268.0 ± 12.1 | 132–134                 | 141.9 ± 5.7  | 2.67–2.92              | 2.83 ± 0.16 | 104–111                 | 118.4 ± 7.3  | McEnroe and Cech (1987)        |
|                  | <i>A. transmontanus</i>      | 3–6 A            | Unk        |           | FW           | BW    | Unk          | Unk   |                      |              | 123–129                 | 130 ± 15.0   | 2.0–2.7                | 2.5 ± 0.4   | 111–116                 | 115.1 ± 7.1  | Unist and Van de Putte (1967)  |
|                  | <i>A. naccarii</i>           | 5 J; 2.5 yr      | 1.5        |           | 0            | 35    | LT           | 20    | ~270                 | ~280         |                         |              |                        |             |                         |              | Martinez-Alvarez et al. (2002) |
| FW               | <i>A. naccarii</i>           | 6 J; 0.4 yr      | 0.06       |           | 0            | 23    | 42           | 42    | 256 ± 5              | 268 ± 3      | 134 ± 4                 | 134 ± 7      | 4.3 ± 0.3              | 5.7 ± 0.2   | 110 ± 2                 | 115 ± 3      | McKenzie et al. (1999)         |
|                  | <i>A. brevirostrum</i>       | 10 J; 1.25 yr    | 0.27       |           | 0            | 20    | 70           | 70    | ~270                 | ~275         | ~125                    | ~125         | ~2.5                   | ~2.1        | ~111                    | ~117         | Jarvis and Ballantyne (2003)   |
|                  | <i>A. brevirostrum</i>       | 6–10 J           | 1.8        |           | FW           | 30    | 25           | 25    | 244.2 ± 1.1          | 312.5 ± 3.3  | 128.9 ± 2.5             | 142.4 ± 1.8  | 2.56 ± 0.19            | 3.25 ± 0.14 |                         |              | Krayushkina (1998)             |
|                  | <i>A. baerii</i> *           | 15 J; 0.6 yr     | 0.12       |           | 0            | 14    | 45           | 45    | ~245                 | ~310         | ~120                    | ~150         | ~2.5                   | ~2.0        | ~110                    | ~150         | Rodriguez et al. (2002)        |
|                  | <i>A. fulvescens</i>         | 5 J; 6 yr        | 1.2        |           | 0            | 25    | LT           | 2     | 246.6 ± 8.9          | 450.8 ± 55.9 | 125.6 ± 4.8             | 223.6 ± 26.0 | 2.4 ± 0.2              | 4.1 ± 0.9   | 116.8 ± 5.8             | 223.1 ± 19.3 | LeBreton and Beamish (1998)    |

Type based on Doroshov (1985). Values are either means ± SE, values provided in paper, or in McEnroe and Cech (1987) and LeBreton and Beamish (1998) values are means ± SD. For this paper combined mean initial weights were used. Unist and Van de Putte (1967) used fish from FW and BW (referred to as SW, but it is actually a BW location), and in Altinok et al. (1998) the SW fish were acclimated to 20‰ for 14 d prior to transfer to 35‰. Abbreviations: Hypo = hypotonic, Hyper = hypertonic, J = juvenile, SA = subadult, A = adult, LT = long term, yr = year, Unk = unknown or not stated, FW = fresh water, BW = brackish water, SW = seawater

\*Some populations may enter BW

energy budget in sturgeon, our results were very similar to those for similar sized Atlantic salmon, *Salmo salar*, at 12°C acclimated to 35‰ and only slightly higher than those acclimated to 0‰ (Maxime et al. 1990).

### Survival and growth

Juvenile green sturgeon survived in all treatments over the duration of this experiment except for the 100 dph fish in SW. The decrease in growth, as assessed by differences in final wet masses, in fish at 100 dph and 170 dph in SW indicates that fish were not adapting well to the hyperosmotic conditions at these ages and sizes. However, at 533 dph, final weights were comparable across salinities, indicating that of the ages and sizes tested, this age group was best adapted to hyperosmotic conditions. Our study could not reveal the actual difference in growth rates due to a lack of replicate tanks and relatively high individual variation.

Salinity effects on growth are complex and may result from action on metabolic rate, food intake, food conversion, and hormonal stimulation (Boeuf and Payan 2001). In our experiment, the effects of salinity on metabolic rate were negligible, and the lack of growth in the 100 dph and 170 dph SW groups may be accounted for by differences in food consumption. Visual observations on food consumption, and estimates of the amount of uneaten food removed from tanks indicated that 100 dph FW and BW fish were eating all or nearly all of their food, whereas the 100 dph SW fish were not. Whereas some of the 100 dph SW fish were feeding vigorously, the majority were not. Aerobic metabolism, as mentioned above, was not different even though fish were starving, probably due to a reallocation of energy expenditure from growth to other processes such as maintenance metabolism (De Boeck et al. 2000). In addition, swimming speeds and activity levels for many fish in the 100 dph SW group appeared to be reduced. Definitive food consumption observations were more difficult in the 170 dph SW group due to a greater depth and volume of water, and periodic turbidity caused by oceanic conditions. Food consumption and swimming activity have been shown to decrease in hyperosmotic salinities in juvenile Siberian

sturgeon (Rodriguez et al. 2002), juvenile lake sturgeon, *A. fulvescens* (LeBreton and Beamish 1998), rainbow trout, *Oncorhynchus mykiss* (McKay and Gjerde 1985), and common carp, *Cyprinus carpio* (Wang et al. 1997; De Boeck et al. 2000). Growth rates in the 170 dph and 533 dph age groups could also have been affected by competition, as the fish from both age groups were kept in the same tanks. Because each treatment group at these ages was subjected to similar potential competition levels, this effect would be expected to be comparable for all treatment groups at each of these ages. Also, the smaller sample sizes for the treatments in the 533 dph fish ( $n = 20$  as compared to 30 for other ages) may have made detection of growth differences at this age more difficult.

Salinity-related effects on growth have also been found in juveniles of other anadromous sturgeon species. For example, Altinok and Grizzle (2001) found that juvenile Gulf sturgeon had higher specific growth rates and higher energy absorption efficiencies in 3‰ and 9‰ than in 0‰ and 1‰, although hyperosmotic salinities were not tested. Jarvis et al. (2001) found that juvenile shortnose sturgeon weight gain and feed conversion efficiency was the highest in FW, and the lowest at 20‰, the highest salinity tested. Juvenile Adriatic sturgeon had lower specific growth rates and food conversion efficiencies at 20‰ than 0‰ (McKenzie et al. 1999) and lower specific growth rates at 11‰ than 0‰ (McKenzie et al. 2001a). Although juveniles of many fish species grow optimally in intermediate salinities (Boeuf and Payan 2001), sturgeon species vary according to life history and associated age/body size differences in osmoregulatory ability.

### Osmoregulation

Several reviews have noted that there appears to be three to four main life history strategies of sturgeons differentiated by whether adults remain in FW (potamodromous), migrate to some body of BW (FW amphidromous and brackish sea migrants), or migrate to full-strength oceanic SW ( $\geq 33$ ‰, anadromous; Doroshov 1985; Rochard et al. 1990; Bemis and Kynard 1997; Billard and Lecointre 2001). Doroshov (1985) termed these

life history strategies as FW, semi-anadromous, and truly anadromous, and for the sake of simplicity, those terms will be used here. Billard and Lecointre (2001) identified six species that utilize full-strength SW (truly anadromous species): Atlantic sturgeon, European sturgeon, Chinese sturgeon, *A. sinensis*, green sturgeon, Sakhalin sturgeon (Asian green sturgeon, *A. mikadoi*; considered a separate species according to Birstein et al. (1993) and Birstein and Bemis (1997)), and white sturgeon. However, Rochard et al. (1990) noted that this was the exception to the rule for white sturgeon, and Doroshov (1985) identified white sturgeon as a semi-anadromous species. Therefore, only five sturgeon species are considered to be truly anadromous, or utilize full-strength SW for a significant portion of their life history. The known osmoregulatory abilities in both hyposmotic and hyperosmotic environments for representatives from each life history strategy are summarized in Table 4.

From a comparison with these other sturgeon species, the results of this study indicate that green sturgeon successfully iono- and osmoregulate at one of the youngest ages known in sturgeons. In our study, we found that green sturgeon iono- and osmoregulatory abilities were clearly age/body size dependent, with osmolality and  $\text{Na}^+$  concentrations highest in SW at 100 dph, and decreasing to FW and BW levels in the 533 dph age group. Furthermore, fish regulated equally well at BW and FW salinities, as indicated by similar osmolality and  $\text{Na}^+$  levels.

In green sturgeon, plasma  $\text{Cl}^-$  levels were always higher in SW, although concentrations decreased with age/size, to levels slightly above FW and BW in the 533 dph groups. Plasma  $\text{Cl}^-$  concentrations are higher in other sturgeon species in hyperosmotic environments as well, even after  $\text{Na}^+$  and  $\text{K}^+$  concentrations decreased to FW levels (McEnroe and Cech 1985; McKenzie et al. 2001a; Jarvis and Ballantyne 2003). Though  $\text{Cl}^-$  was not regulated to FW or BW levels, concentrations were similar to or slightly lower than plasma  $\text{Na}^+$  concentrations in this experiment.

Plasma  $\text{K}^+$  was fairly similar between salinities, with small differences seen at 100 dph and 533 dph. Interestingly, plasma  $\text{K}^+$  has not been shown to change between many FW and

hyperosmotic acclimated sturgeon, regardless of their size (Altinok et al. 1998; Rodriguez et al. 2002; Jarvis and Ballantyne 2003). Perhaps plasma  $\text{K}^+$  is more tightly regulated due to its important role in iono- and osmoregulation such as in  $\text{Na}^+$ ,  $\text{K}^+$ -ATPase driven ionic and concentration gradients and water uptake mechanisms.

Results of this study are not directly comparable to a previous experiment on subadult-adult (6.0–9.6 kg) green sturgeon by Potts and Rudy (1972) because of insufficient acclimation time for ion homeostasis following salinity transfer (Table 4). Potts and Rudy (1972) evaluated osmoregulatory abilities in FW and SW after 3 d, following a 2-d gradual transfer from SW or 60 % SW (ca. 20‰). Their mean  $\text{Na}^+$ ,  $\text{K}^+$ , and  $\text{Cl}^-$  values in FW are well below the FW plasma values from our experiment. They likely reflect transient ion losses associated with acute transfer to FW. Their mean values of  $\text{Na}^+$  and  $\text{K}^+$  in SW fish were below most FW values, and all BW and SW values from our experiment. Their  $\text{Cl}^-$  values were most similar to the SW 170 dph group from this experiment, and above those in the SW 533 dph group. The transfer from BW to SW may not have represented a large osmotic challenge to their comparatively larger fish because the gills would have already supported cellular changes necessary to osmoregulate in hyperosmotic salinities (Allen 2005). Therefore, the larger body sizes of the green sturgeon used by Potts and Rudy (1972) explain why the ionic constituents were regulated at a lower homeostatic level, compared with our juvenile fish.

Other important indicators of iono- and osmoregulatory abilities: plasma glucose, plasma lactate, plasma proteins, hematocrit, and muscle water content also show that juvenile green sturgeon can successfully transition to SW by 533 dph. Plasma glucose, lactate, and protein concentrations were all low in SW as compared to FW and BW treatment groups at 100 dph and 170 dph, but were similar by 533 dph. The decreased concentrations in SW at 100 dph and 170 dph were a result of reduced food consumption. Many fish in the 100 dph group, which experienced 23% mortality, had stopped feeding, therefore not replenishing energy stores and fueling osmoregulatory costs. Results in the

170 dph and 533 dph groups could have been affected by potential competitive feeding interactions, as mentioned previously.

Lactate increases in the plasma due to glycolysis in white muscle (Driedzic and Hochachka 1978) following periods of hypoxia or bouts of high activity McKenzie et al. (2001a; Baker et al. 2005). Our fish were not exposed to hypoxia, but low lactate concentrations did appear to be related to swimming activity. Plasma lactate was lowest at 100 dph and 170 dph in SW treatments, and swimming activity also appeared to be reduced in the majority of these fish, although observation in the SW treatment at 170 dph was more difficult due to tank dimensions and periodic water turbidity due to ocean conditions. The decreased activity in these two groups was most likely due to decreased food consumption and energy stores. Reduced swimming activity at high salinities has been found in other juvenile sturgeons (LeBreton and Beamish 1998), although Jarvis and Ballantyne (2003) found no differences in juvenile shortnose sturgeon activity between hyposmotic and hyperosmotic salinities despite reduced growth rates. In our study, the low plasma lactate concentrations at 100 dph and 170 dph in SW were similar to previously reported resting values in green sturgeon acclimated to 11°C at 0‰ (Lankford et al. 2003), whereas concentrations in other groups were higher probably indicating moderate activity levels, but lower than those measured in exhaustively exercised green sturgeon (Warren et al. 2004).

Plasma proteins, and in particular albumin which can make up >20% of the total plasma proteins in sturgeons (Khabarov et al. 2002), help to regulate fluid balance through maintenance of oncotic pressure in systemic vasculature. Plasma protein concentrations can decrease immediately after hyperosmotic exposure, possibly due to an ion-induced increase in extracellular volume (Nichols & Weisbart 1985). However, plasma proteins have been found to be positively correlated to salinity after a longer-term acclimation period in gilthead sea bream, *Sparus aurata* (Sangiao-Alvarellos et al. 2003), and tilapia, *O. niloticus* × *O. mossambicus* (Verdegem et al. 1997). The low plasma proteins in the SW 100 dph and 170 dph groups may stem from

regulatory mechanisms associated with the high ionic concentrations in the plasma or from the observed lack of foraging. Plasma proteins have been found to be related to the nutritional status of the fish, increasing with higher ingested protein concentrations and food quantity (Verdegem et al. 1997; Alcorn et al. 2003). They may also provide an energy source during starvation periods via gluconeogenic pathways. In our experiment, the comparable concentrations of plasma proteins in all treatments at 533 dph suggests more effective ionoregulation and more frequent feeding in SW at this age. Plasma proteins were also reduced in juvenile Adriatic sturgeon after 20 d at 35‰ (Martinez-Alvarez et al. 2002). However, plasma proteins were not different from freshwater concentrations in juvenile shortnose (Jarvis and Ballantyne 2003) or Siberian (Rodriguez et al. 2002) sturgeons after long-term acclimation to 20‰ and 14‰ respectively.

Hematocrit can increase in stressful circumstances (Soivio and Nikinmaa 1981). Jarvis and Ballantyne (2003) suggested that hematocrit levels rose in juvenile shortnose sturgeon after 10 weeks in hyperosmotic conditions as a result of stress. Because hematocrit changed little between salinities and ages in our experiment, it was not a good stress indicator in juvenile green sturgeon, particularly when compared with osmolality and growth. Similarly, no differences were found in hematocrit between long-term FW and SW acclimated juvenile Adriatic (Martinez-Alvarez et al. 2002) or Gulf (Altinok et al. 1998) sturgeons, and juvenile Atlantic and shortnose sturgeons exercised vigorously (Baker et al. 2005).

Muscle water typically declines immediately after hyperosmotic exposure, and then is regulated back to original FW levels after a period of time in anadromous salmonids (Handeland et al. 1998). However, the prolonged high muscle water contents in the 100 dph and 170 dph SW groups were likely due to muscle catabolism to supply the energy demands after many fish had stopped feeding (Heap et al. 1986; McLeese and Moon 1989; Maddock and Burton 1994). In starved fish in SW, muscle water also appears to increase as a result of Na<sup>+</sup> and Cl<sup>-</sup> increases in the extracellular spaces, which osmotically draws in more water (Love et al. 1968; Sutton 1968, but see MacLeod

1977). The fact that the SW group at 533 dph was similar to FW and BW muscle water contents, indicates that at this age, the fish were feeding and osmoregulating sufficiently. Altinok et al. (1998) and Martinez-Alvarez et al. (2002) both found that muscle water content fell initially in juvenile Gulf and Adriatic sturgeons after hyperosmotic exposure, but was similar to FW levels by the end of the experiments. Similarly, Jarvis and Ballantyne (2003) found muscle moisture was not different between juvenile shortnose sturgeon acclimated to FW and 20‰, although white muscle lipids (not measured in our experiments) appeared to be important for supplying the energy needed during salinity challenge.

Salinity acclimation studies in sturgeons show that ionic and osmoregulatory abilities are positively correlated with fish body size (McEnroe and Cech 1985, 1987; Altinok et al. 1998; LeBretton and Beamish 1998; Cataldi et al. 1999). Juveniles have higher osmolality and plasma ion levels than adults of the same species in the same salinities (McEnroe and Cech 1985, 1987; Table 4). Presumably, because sturgeons hatch and grow at least initially in FW, they regulate at elevated ion homeostasis levels in hyperosmotic salinities, as compared to hyposmotic salinities (McEnroe and Cech 1985, 1987; Altinok et al. 1998; Krayushkina 1998; McKenzie et al. 1999; Rodriguez et al. 2002). Some juveniles also appear to regulate at nearly isosmotic levels, when salinities are approximately 8–12‰ (Natochin et al. 1985; Shelukhin et al. 1990). In contrast, adults, which are larger, are more efficient ion regulators, maintaining osmolality and ions at similar concentrations between hyperosmotic and hyposmotic environments (Urist and Van de Putte 1967; McEnroe and Cech 1985, 1987). However, if acclimation time is limited (e.g., 3 d), osmolality and ion concentrations are still elevated in hyperosmotic environments (Potts and Rudy 1972; McEnroe and Cech 1985).

Sturgeon species osmoregulate similarly to teleosts, in that they are hyperosmotic with respect to FW and hyposmotic with respect to SW. Plasma osmolality and electrolyte concentrations of juvenile green sturgeon and other truly anadromous sturgeon species in FW and SW (Table 4) are comparable or slightly lower than typical

ranges for teleosts in FW and SW (Holmes and Donaldson 1969). Anadromous teleosts, notably salmonids, undergo a parr-smolt transformation that is orchestrated by neuroendocrine regulation and rapidly prepares the fish for life in SW (Hoar 1988). Though this hormone-mediated transformation has not been demonstrated in sturgeons, studies have shown that salinity tolerance is improved after acclimation to mildly hyperosmotic salinities (McEnroe and Cech 1985; Altinok et al. 1998; McKenzie et al. 2001b). This acclimation period may be needed to initiate enzymatic and cellular osmoregulatory changes necessary for osmoregulation in higher salinities. Available field evidence suggests juvenile anadromous and semi-anadromous sturgeons remain in low salinity habitats, presumably entering hyperosmotic salinities such as SW after several years (Natochin et al. 1985; Rochard et al. 2001, review by Wilson and McKinley 2004). Because studies also indicate a clear effect of body size on osmoregulation at hyperosmotic salinities, sturgeons need to be larger to osmoregulate at higher salinities, such as those in SW, compared to anadromous teleosts. Potts and Rudy's (1972) limited data suggest that sturgeon may not be as efficient of iono- and osmoregulators as are anadromous teleosts. They found that a green sturgeon ( $n = 1$ ) transferred from hyperosmotic salinities to freshwater took longer to slow the rate of ion loss than is typical in teleosts. Perhaps salinity transition in sturgeons requires more time to reach ionic homeostasis, compared to anadromous teleosts.

The green sturgeon's "strategy" appears to maximize early entry to SW through large initial size of larva and rapid growth of juveniles (Deng et al. 2002), thereby reaching the appropriate size to enter SW as soon as possible. Acquisition of SW tolerance at an early age is especially important in some of their natal rivers, such as the Klamath, which has a high gradient and a small estuary. Limited field results have found fish from 1 to 4 years old in estuaries, with very few fish from 3 to 13 years of age and 70–115 cm TL,<sup>3</sup> which corresponds closely to the mean size of fish at 533 dph in our experiments (75 cm, 1.5 kg). This field data

<sup>3</sup> Ibid.

and our results also suggest that some individuals may emigrate to SW within a year while others may remain in the estuary for up to 4 years.

## Conclusion

One of the most relevant questions to ask when determining where juvenile green sturgeon will be found at different ages and sizes, is are they thriving or just surviving? Based on our data, all age groups of sturgeon did well in FW and BW. The majority of fish at 100 dph survived in SW, although many died or were in poor condition. However, approximately 20% appeared to be osmoregulating and feeding. All other age and salinity groups had no mortality. Osmoregulatory costs between salinities were not great enough to result in differences in aerobic metabolic rates, when the fish were properly acclimated to the test salinity. Final wet masses were depressed in SW compared to FW and BW, except at 533 dph, and were due, at least in part, to reduced or cessation of food consumption. Energy stores were depleted in 100 and 170 dph SW groups, as evidenced by low plasma glucose levels, low plasma protein concentrations, and high muscle water content, indicating that the fish were likely starving. Therefore, even though the whole-animal cost of salinity exposure did not affect aerobic metabolic costs, remaining energy stores were being slowly used up through tissue catabolism. Osmoregulatory abilities were clearly best at 533 dph in SW based on plasma osmolality, ions, glucose, lactate, protein concentrations, and muscle water contents, which were at similar levels to FW and BW groups. Therefore, most juvenile green sturgeon are likely to be found in FW to BW habitats at least up to 170 dph (6 months) and 0.15 kg wet body mass, and are able to move into SW around 533 dph (1.5 years) and 1.5 kg (75 cm). The limited field data on the North American green sturgeon and on its Asian form, Sakhalin sturgeon, indicate that this age and body size is near the size at which sturgeon are no longer found in FW or BW habitats<sup>4</sup> (Andrey Shmigrilov

personal communication). However, there is likely some population heterogeneity, with some sturgeon in our experiments as young as 100 dph able to successfully make the transition to SW, suggesting that emigration to SW may occur in a wider range of age intervals.

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<sup>4</sup> Ibid.

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